Wisconsin Breast Cancer Diagnosis (B vs M) —

Multivariate Classification & Interpretation

SEP767 - Final Project

Andi Dong

“Can unsupervised compression compete with supervised latent modeling?”

Title & Problem Context

- *Significance: [Accurate and interpretable early screening modelscan support clinical decision-making by reducing missedmalignant cases and improving diagnostic consistency.]*
- *Dataset: Wisconsin Diagnostic dataset(from Kaggle)*
- *Goal: high sensitivity for malignant detection + interpretable features*
- *Overview of methods: PCA+RF baseline(CV) + PLS-DA (proper validation) + VIP + multiblock interpretation*

PIPELINE

Data → Preprocess → Models → Evaluation → Interpretation

Related Work: PCA and PLS-DA in Biomedical Classification

- **PCA-based dimensionality reduction** has been widely adopted in cancer-related high-dimensional datasets to reduce redundancy and improve classification robustness. Studies on microarray cancer data demonstrate that PCA significantly improves classification accuracy by addressing the imbalance between feature dimensionality and sample size*(Adiwijaya et al., 2018)*
- **Multivariate statistical methods**, including PCA and PLS-based approaches, have shown strong performance in breast cancer diagnosis. Compared to univariate analysis, multivariate models achieve higher accuracy and better class separability in complex biomedical data*(Kim et al., 2009)*
- **PLS-DA**, as a supervised multivariate technique, has been extensively applied in biomedical classification tasks. Comparative studies indicate that PLS-DA achieves high accuracy, sensitivity, and specificity (typically above 90%) in high-dimensional and highly correlated biomedical datasets*(Lasalvia et al., 2022)*

Database Description

- Samples: **569 cell nucleus**
- Features: **30 numeric features** after dropping id/empty column

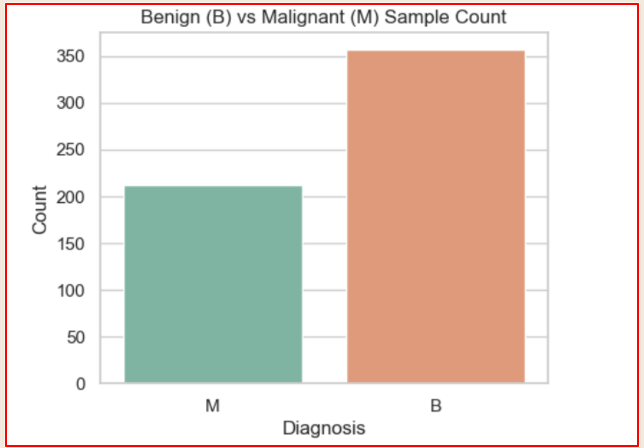
```
Step 1 data preprocessing! X_scaled : (569, 30)    y : (569,)
```

- Labels: **B=357, M=212 (Diagnosis (M = malignant, B = benign))**
- Label encoding: **B→0, M→1**

a) radius (mean of distances from center to points on the perimeter)
b) texture (standard deviation of gray-scale values)
c) perimeter
d) area
e) smoothness (local variation in radius lengths)
f) compactness (perimeter^2 / area - 1.0)
g) concavity (severity of concave portions of the contour)
h) concave points (number of concave portions of the contour)
i) symmetry
j) fractal dimension ("coastline approximation" - 1)

The mean, standard error and "worst" or largest (mean of the three largest values) of these features were computed for each image,

```
diagnosis
B    357
M    212
Name: count, dtype: int64

映射后的标签统计:
diagnosis_num
0    357
1    212
Name: count, dtype: int64
```



Objectives

- Build a baseline classifier using **PCA + Random Forest**
- Build a rigorously validated **PLS-DA** model (train/CV, test threshold, validation final)
- Provide interpretability via **VIP** + **multiblock PCA** (mean/se/worst)

Preprocessing & PCA --- Methodology

- Drop non-informative columns (id, empty)
- Encode label, split X/y
- Standardize features (z-score) before PCA/PLS
- PCA for visualization + dimension reduction

```python
# delete id and Unnamed: 32 if exists
cols_to_drop = []

if "id" in data.columns:
    cols_to_drop.append("id")
if "Unnamed: 32" in data.columns:
    cols_to_drop.append("Unnamed: 32")

data = data.drop(columns=cols_to_drop)
```

```python
# ====== split X y ======
# feature columns: all numerical features except diagnosis and diagnosis_num
feature_cols = [col for col in data.columns
                if col not in ["diagnosis", "diagnosis_num"]]

X = data[feature_cols]
y = data["diagnosis_num"]
```

```python
# ====== standardized features ======
scaler = StandardScaler()

# 注意：fit_transform 返回的是 numpy 数组
X_scaled_array = scaler.fit_transform(X)
```

```python
# ====== Run PCA ======
pca = PCA()
pca.fit(X_scaled)

# 获取PCA后的样本坐标 (scores)
X_pca = pca.transform(X_scaled)
```

Two modeling tracks --- Methodology

- Track A: PCA + RF (baseline)
  - Use first PCs as inputs (baseline used PC1–PC3)
  - Report confusion matrix and accuracy
  - Then select PC number via Pipeline + 5-fold CV, choose best PCs (best=5)

```python
# ====== 划分训练集和测试集 ======
# test_size=0.2 表示 20% 作测试，随机数种子设置为 42，保证结果可复现
X_train, X_test, y_train, y_test = train_test_split(
    X_pca_3, y, test_size=0.2, random_state=42
)

# ====== 定义 Random Forest 模型 ======
rf = RandomForestClassifier(
    n_estimators=200,      # 决策树数量
    max_depth=None,        # 不限制树深度
    random_state=42
)

# ====== 训练模型 ======
rf.fit(X_train, y_train)
```

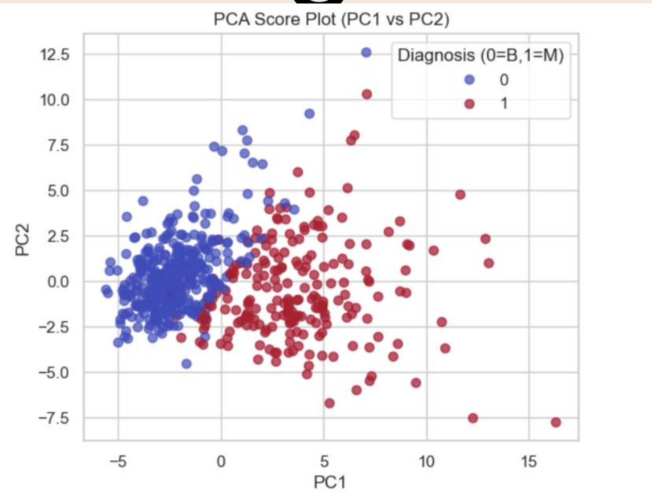
- Track B: PLS-DA (golden rules)
  - Train (CV select LV by ROC-AUC)
  - Test (tune threshold using Youden's J)
  - Validation (final metrics only)

```python
scaler = StandardScaler()
X_train_s = scaler.fit_transform(X_train)
X_test_s  = scaler.transform(X_test)

pls = PLSRegression(n_components=best_lv)
pls.fit(X_train_s, y_train)

y_test_score = pls.predict(X_test_s).ravel()
```

My using rules!

- Stratified Cross-Validation: StratifiedKFold was used due to the imbalanced sample sizes between benign and malignant classes.
- Decision Threshold Selection: The decision threshold on the testing set was determined using Youden's J, rather than a fixed threshold of 0.5.
- Final Model Training: For final validation, the model was retrained using the combined training and testing datasets.

PCA insight --- Results



first two PC loadings:

	PC1	PC2
radius_mean	0.218902	-0.233857
texture_mean	0.103725	-0.059706
perimeter_mean	0.227537	-0.215181
area_mean	0.220995	-0.231077
smoothness_mean	0.142590	0.186113

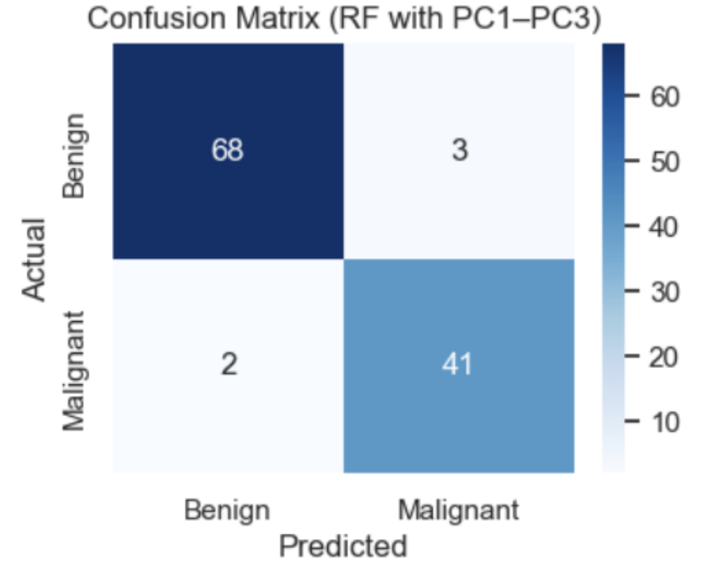
- a clear separation trend between benign and malignant tumors along PC1
- PC1 is dominated by size-related features (radius, perimeter, area)

PCA+RF baseline + CV selection --- Results

Using 3 components as a base line

```
accuracy = accuracy_score(y_test, y_pred)
print("Random Forest Accuracy (using PC1-PC3): {:.4f}".format(accuracy))
```

```
Random Forest Accuracy (using PC1-PC3): 0.9561
```

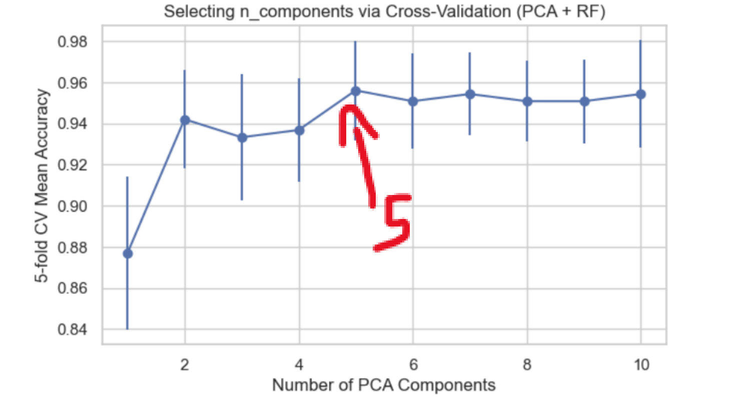


Using Cross-Validation to find the optimum # of components

	n_components	cv_mean_accuracy	cv_std
0	1	0.876929	0.037406
1	2	0.942059	0.023865
2	3	0.933271	0.030626
3	4	0.936796	0.024989
4	5	0.956094	0.024149
5	6	0.950831	0.023229
6	7	0.954339	0.020264
7	8	0.950831	0.019639
8	9	0.950815	0.020417
9	10	0.954324	0.026233

PC # = 5

```
最佳PC数量（CV mean最高）: 5
对应CV accuracy: 0.9561 ± 0.0241
```

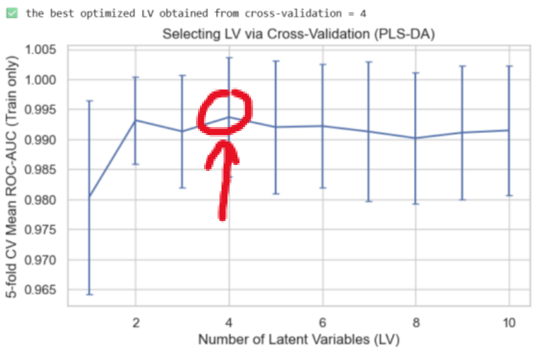


PLS-DA validation performance --- Results

60/20/20 : Train 60%, Test 20%, Validation 20%

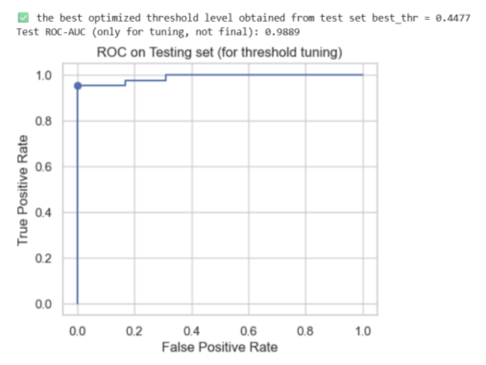
```
Train: (341, 30) Test: (114, 30) Validation: (114, 30)
Train class ratio: {0: 0.628, 1: 0.372}
Test  class ratio: {0: 0.623, 1: 0.377}
Val   class ratio: {0: 0.632, 1: 0.368}
```

The best LV obtained from CV = 4 based on AUC



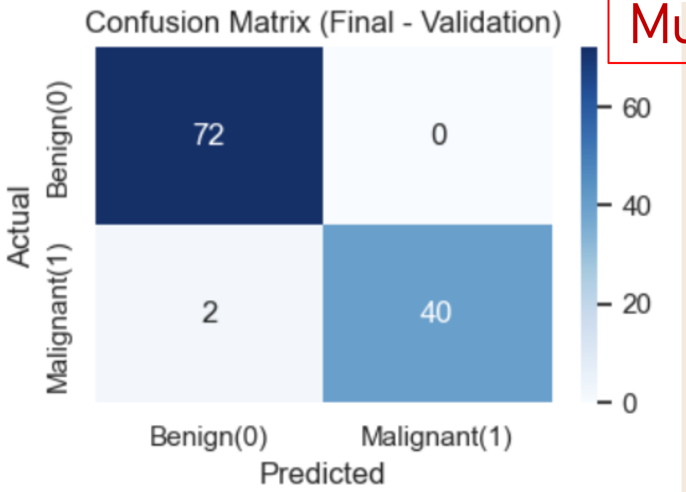
	lv	cv_mean_auc	cv_std_auc
0	1	0.980291	0.016128
1	2	0.993170	0.007218
2	3	0.991311	0.009324
3	4	0.993680	0.009959
4	5	0.992030	0.010999
5	6	0.992205	0.010311
6	7	0.991285	0.011626
7	8	0.990193	0.010966
8	9	0.991114	0.011156
9	10	0.991481	0.010791

The best threshold obtained from test set =0.4477



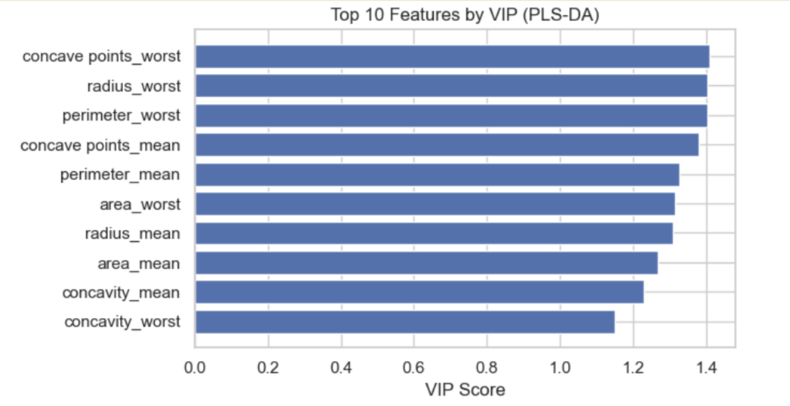
Results from the final validation set

```
================ FINAL (Validation Set Only) ================
Confusion Matrix:
 [[72  0]
 [ 2 40]]
Accuracy    = 0.9825
Sensitivity = 0.9524  (恶性检出率，越高越少漏检)
Specificity = 1.0000  (良性识别率)
ROC-AUC     = 0.9944
PR-AUC      = 0.9924
```



Much Better Result!

Interpretation (VIP + “what drives malignancy”) (Discussion)



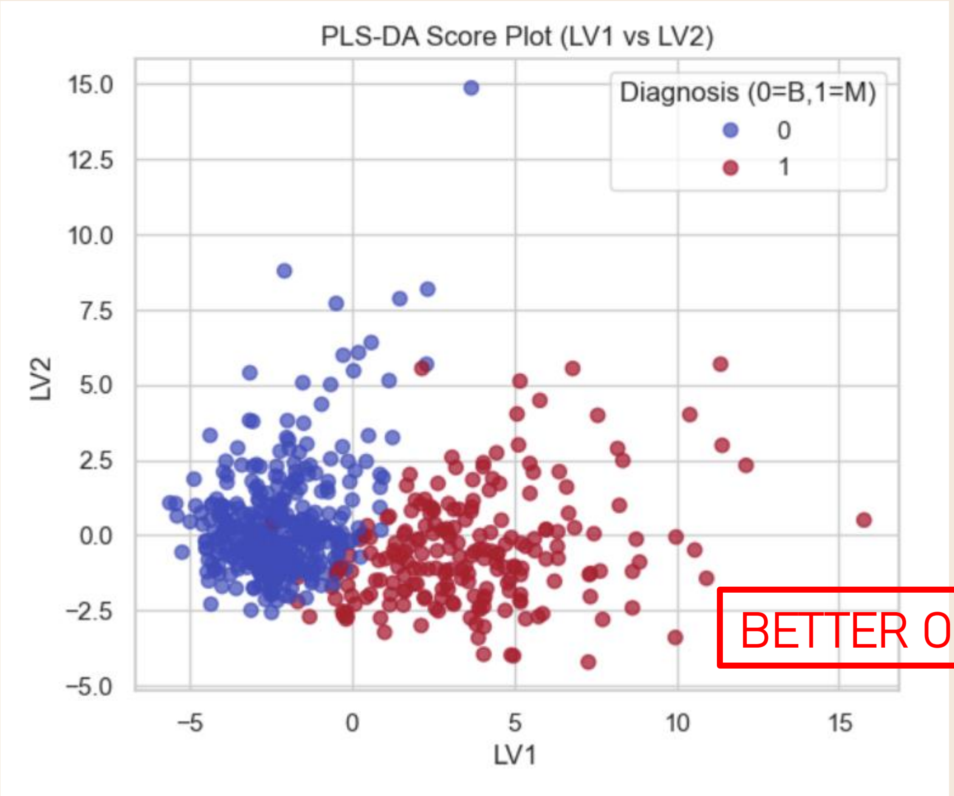

```
Top 10 VIP features:
```

	feature	VIP
27	concave points_worst	1.409697
20	radius_worst	1.404007
22	perimeter_worst	1.403902
7	concave points_mean	1.380496
2	perimeter_mean	1.327158
23	area_worst	1.315342
0	radius_mean	1.310553
3	area_mean	1.267173
6	concavity_mean	1.230833
26	concavity_worst	1.149302

- The worst concave points feature is the most discriminative variable
- Highlighting the importance of extreme morphological irregularity in malignant tumors.
- The discrimination between benign and malignant tumors is mainly driven by extreme geometric irregularities and worst-case morphology

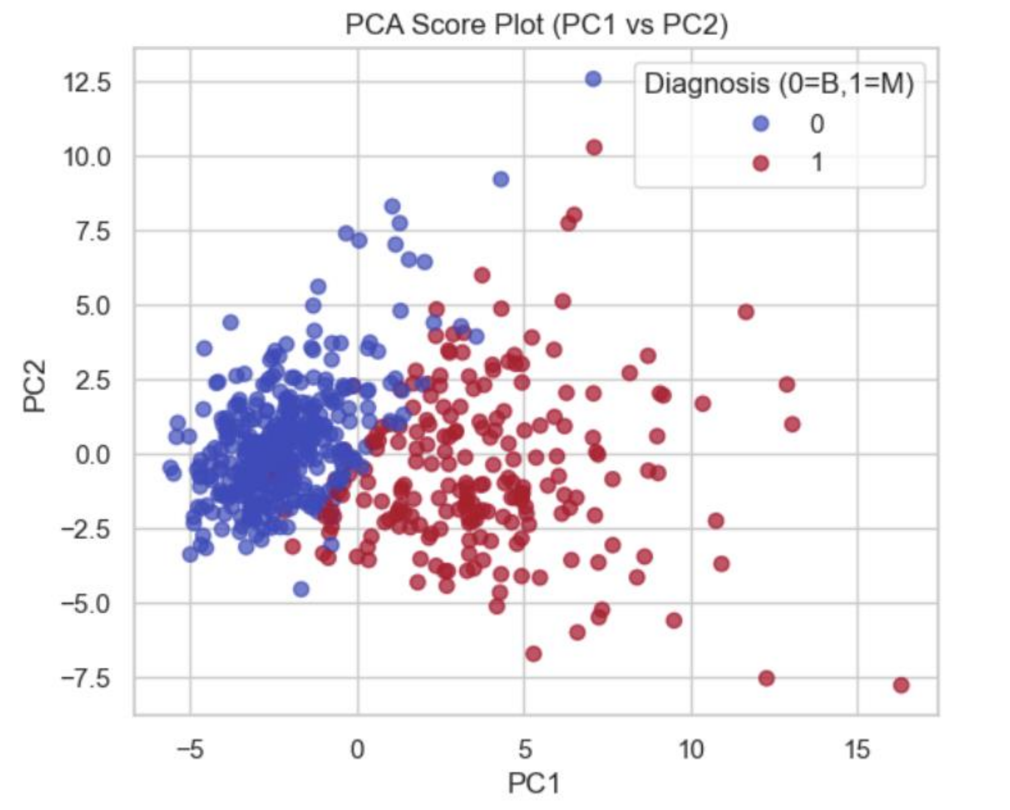
PLS-DA vs PCA under
of Components = 2(use for comparison)

PLS-DA(trained with final validation set using full X)



BETTER ONE!

PCA(using full X)



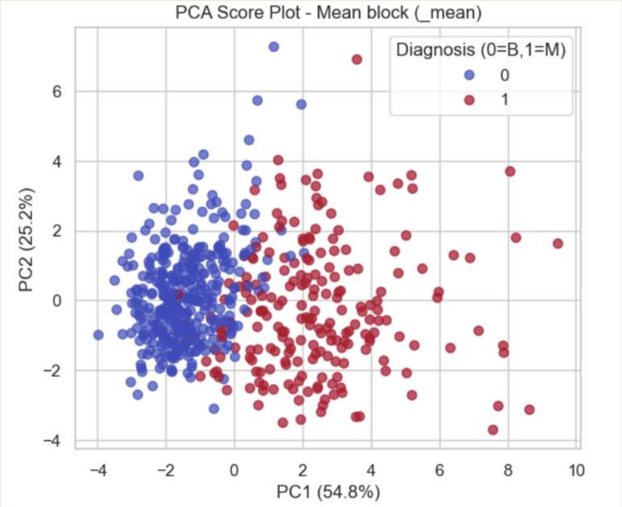
Multiblock Analysis (Discussion Extension)

- Block 1 --> _mean(10)
- Block 2 --> _se(10)
- Block 3 --> _worst(10 vars)

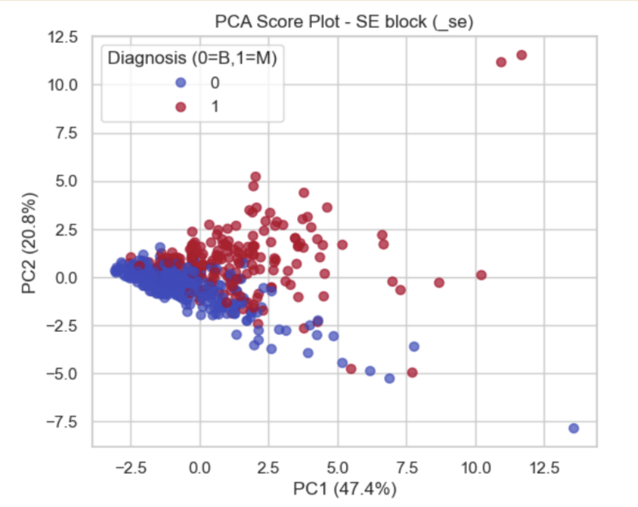
✓ Block with greatest separation from PC1?

✓ Variables with greatest loadings in the block?

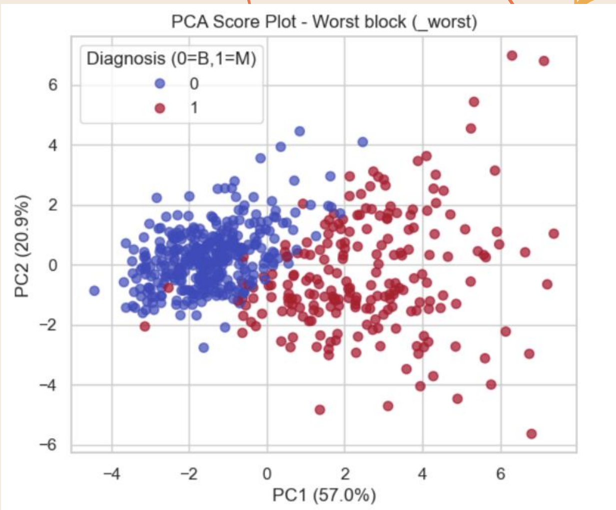
```
Block sizes: {'Mean block (_mean)': 10, 'SE block (_se)': 10, 'Worst block (_worst)': 10}
```




```
=== Mean block (_mean) ===
Top PC1 loadings:
concave points_mean     0.418038
concavity_mean          0.395748
perimeter_mean          0.376044
compactness_mean        0.364442
area_mean               0.364086
dtype: float64
Top PC2 loadings:
fractal_dimension_mean     0.571768
smoothness_mean            0.401962
symmetry_mean              0.368301
radius_mean               -0.313929
area_mean                 -0.304842
dtype: float64
```




```
=== SE block (_se) ===
Top PC1 loadings:
concave points_se     0.385743
compactness_se        0.374799
perimeter_se          0.357481
concavity_se          0.355553
radius_se             0.345592
dtype: float64
Top PC2 loadings:
area_se                  0.500211
radius_se                0.440354
perimeter_se             0.420303
fractal_dimension_se    -0.352502
smoothness_se           -0.270953
dtype: float64
```




```
=== Worst block (_worst) ===
Top PC1 loadings:
concave points_worst     0.397637
concavity_worst          0.374742
compactness_worst        0.364568
perimeter_worst          0.348151
radius_worst             0.335[illegible]10
dtype: float64
Top PC2 loadings:
fractal_dimension_worst     0.[illegible]7849[illegible]
area_worst                 -0.[illegible]
radius_worst               -0.403137
perimeter_worst            -0.375518
smoothness_worst            0.337870
dtype: float64
```

The best Separation

Conclusions & Future Work

Conclusion

- PCA reveals a clear separation trend along PC1; size/morphology features dominate.
- PCA+RF baseline achieved Acc ≈ 0.956; CV suggested best PCs = 5.
- Properly validated PLS-DA achieved Acc = 0.9825, Sens = 0.9524, Spec = 1.0000 (Validation only).
- VIP & multiblock consistently indicate worst-case morphology (e.g., concave points_worst) drives malignancy separation.

Future Work

- Use nested CV and compare with alternative classifiers (logistic/SVM) on PCA or PLS scores.
- Optimize threshold for clinical preference (cost-sensitive, prioritize Sensitivity).
- External validation / larger dataset; add calibration and confidence intervals (bootstrap).

Reference

1. *Adiwijaya, W. U., Lisnawati, E., Aditsania, A., & Kusumo, D. S. (2018). Dimensionality reduction using principal component analysis for cancer detection based on microarray data classification. Journal of Computer Science, 14(11), 1521-1530.*
2. *Lasalvia, M., Capozzi, V., & Perna, G. (2022). A Comparison of PCA-LDA and PLS-DA Techniques for Classification of Vibrational Spectra. Applied Sciences, 12(11), 5345. https://doi.org/10.3390/app12115345*
3. *Kim, Y., Koo, I., Jung, B. H., Chung, B. C., & Lee, D. (2009, November). Multivariate classification of urine metabolome profiles for breast cancer diagnosis. In Proceedings of the third international workshop on Data and text mining in bioinformatics (pp. 11-18).*
4. *[UCI Machine Learning].([Sep 25, 2016]).[Breast Cancer Wisconsin (Diagnostic) Data Set],[Version 2]. Retrieved [Dec 15, 2025] from [Breast Cancer Wisconsin (Diagnostic) Data Set].*

Thank you!

ANDI DONG

Donga29@mcmaster.ca